Keep all tiles flat and in one piece, joined along whole sides without overlaps. Count every side with no tile beside it, including sides around holes. Turns and flips count as the same shape.

Problem 1: Use eight tiles to make a shape with the shortest boundary you can and one with the longest boundary you can. Find a different shape reaching each of those lengths.



Shortest boundary: _____



Longest boundary: _____



Shortest boundary: _____



Longest boundary: _____

Problem 2: Add one tile to each shape. Find every different change you can make to its boundary length.

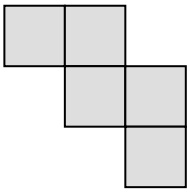


Before: _____

After: _____
Change: _____

After: _____
Change: _____

After: _____
Change: _____

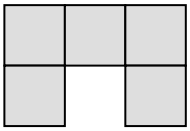


Before: _____

After: _____
Change: _____

After: _____
Change: _____

After: _____
Change: _____

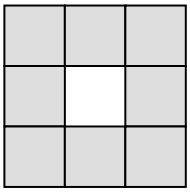


Before: _____

After: _____
Change: _____

After: _____
Change: _____

After: _____
Change: _____



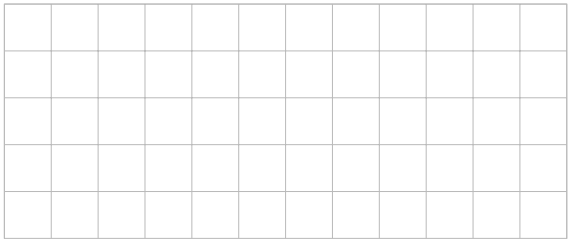
Before: _____

After: _____
Change: _____

After: _____
Change: _____

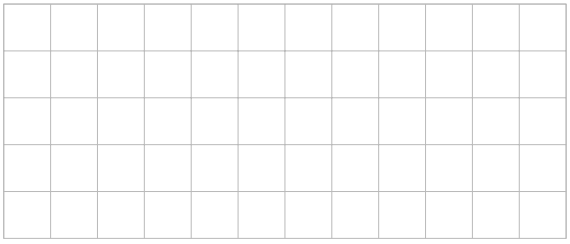
After: _____
Change: _____

Problem 3: A shared side is where two tiles meet. Use ten tiles to make shapes with 9, 10, 11, 12, and 13 shared sides. Find a rule that gives the boundary length from the number of tiles and the number of shared sides.



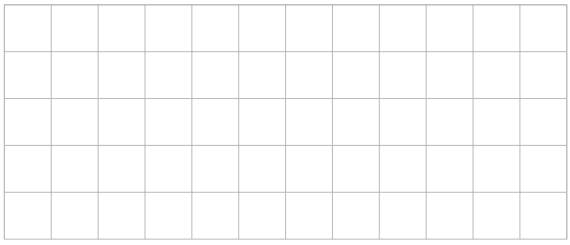
9 shared sides

Boundary: _____



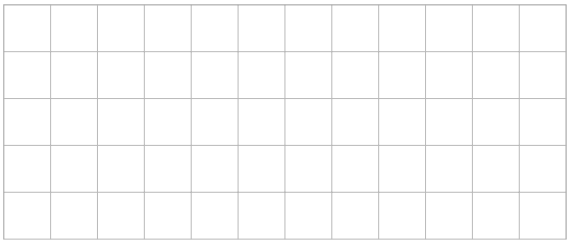
10 shared sides

Boundary: _____



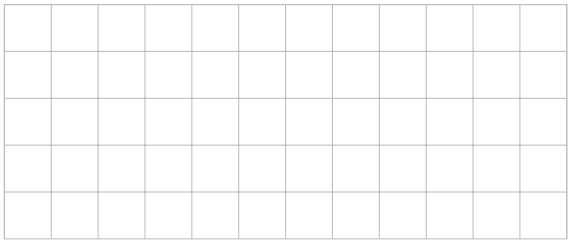
11 shared sides

Boundary: _____



12 shared sides

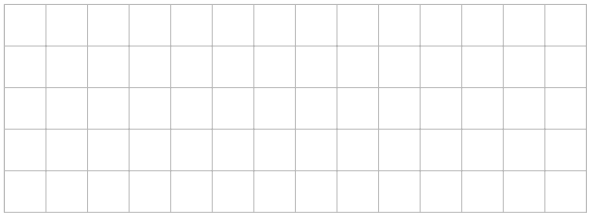
Boundary: _____



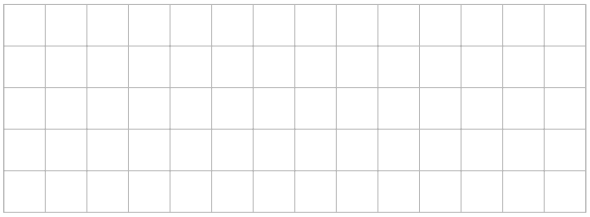
13 shared sides

Boundary: _____

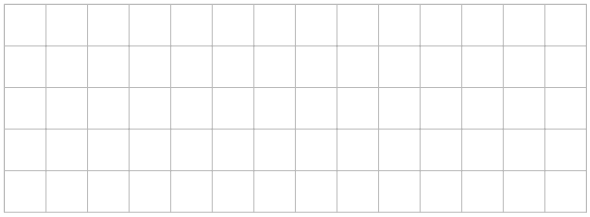
Problem 4: Find the longest possible boundary with 4, 7, 10, and 12 tiles. Explain why no shape with that many tiles can have a longer boundary.



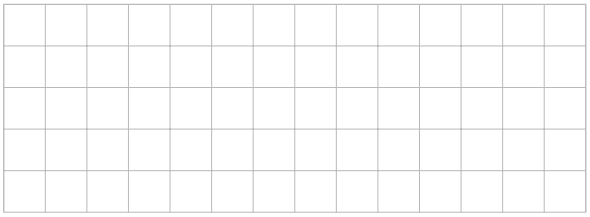
4 tiles; boundary: _____



7 tiles; boundary: _____



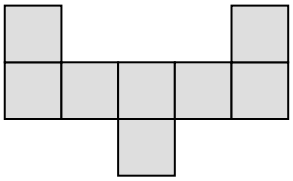
10 tiles; boundary: _____

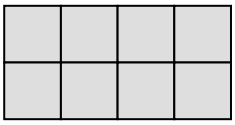


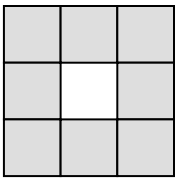
12 tiles; boundary: _____

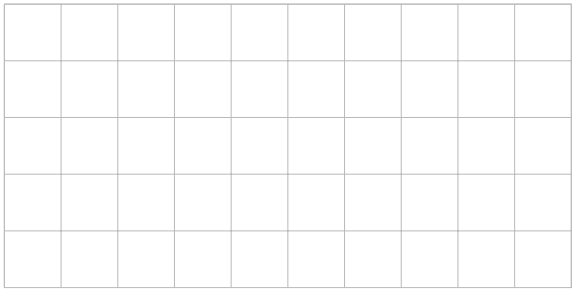
Problem 5: Which of these eight-tile shapes have the longest possible boundary? Explain why each chosen shape reaches that length. Draw two more such shapes, with no straight row or column of four tiles.

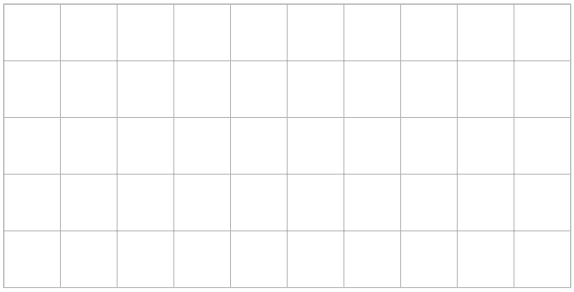




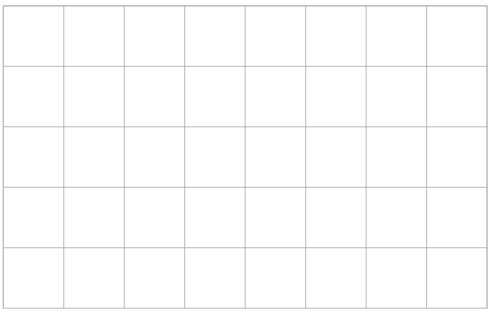






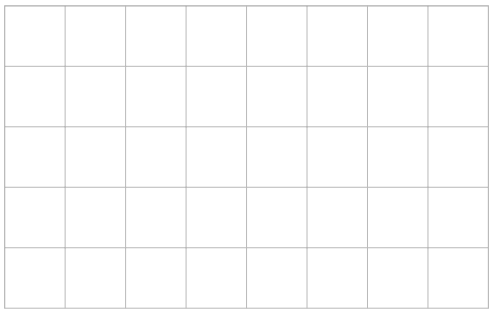


Problem 6: Add one tile to a shape so that its boundary changes by each amount shown. For each amount, use as few starting tiles as possible. Explain why fewer tiles cannot work.



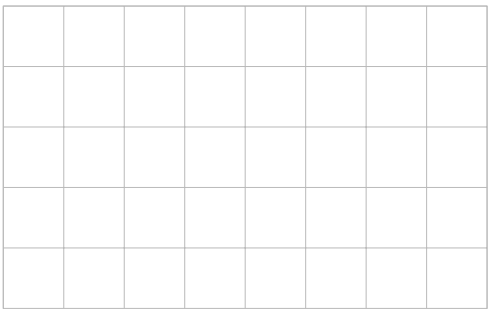
Boundary change: $+2$

Starting tiles: _____



Boundary change: 0

Starting tiles: _____



Boundary change: -2

Starting tiles: _____



Boundary change: -4

Starting tiles: _____